

Ziyang Guo

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Education

- 2022–27* **Ph.D. in Computer Science**
Chair: Jessica Hullman
Northwestern University, Evanston, IL
- 2018–22 **B.E. in Computer Science**
Honors with distinction.
Zhejiang University, Hangzhou, China

Awards & Honors

- 2019 Chu Kochen Honors College Innovation Scholarship (top 2%)
2019 Zhejiang Provincial Scholarship (top 5%)

Publications

Peer-reviewed Conference Proceedings

- 2026 **ComplLLM: Fine-tuning LLMs to Discover Complementary Signals for Decision-making**
Guo, Z. and Wu, Y. and Hartline, J. and Holstein, K. and Hullman, J.
Neural Information Processing Systems (NeurIPS '26)
- 2026 **Explanations are a Means to an End: Decision Theoretic Explanation Evaluation**
Guo, Z., Ustun, B., Hullman, J.
Forty-Third International Conference on Machine Learning (ICML)
- 2026 **The Value of Information in Human-AI Decision-making**
Guo, Z. and Wu, Y. and Hartline, J. D. and Hullman, J.
The Fourteenth International Conference on Learning Representations (ICLR)
- 2024 **A Decision Theoretic Framework for Measuring AI Reliance**
Guo, Z., Wu, Y., Hartline, J. D., and Hullman, J.
ACM Conference on Fairness, Accountability, & Transparency (Proceedings of FAccT 2024)
- 2020 **Visual Analytics of Multivariate Event Sequence Data in Racquet Sports**

*Expected.

Wu, J., **Guo, Z.**, Wang, Z. and Xu, Q. and Wu, Y.
2020 IEEE conference on visual analytics science and technology (VAST)

Journal Articles

- 2024 **VMC: A Grammar for Visualizing Statistical Model Checks.**
Guo, Z., Kale, A., Kay, M., Hullman, J.
IEEE Transactions on Visualization and Computer Graphics (Proceedings of IEEE
VIS 2024, 23.2%, 129/557)
- 2023 **Evm: Incorporating Model Checking into Exploratory Visual Analysis.**
Kale, A., **Guo, Z.**, Qiao, X. L., Heer, J., and Hullman, J.
IEEE Transactions on Visualization and Computer Graphics (Proceedings of IEEE
VIS 2023, 24.6%, 133/539)
- 2023 **The Rational Agent Benchmark for Data Visualization.**
Wu, Y., **Guo, Z.**, Mamakos, M., Hartline, J., and Hullman, J.
IEEE Transactions on Visualization and Computer Graphics (Proceedings of IEEE
VIS 2023, 24.6%, 133/539)
- 2022 **Rasipam: Interactive Pattern Mining of Multivariate Event Sequences in Racket Sports.**
Wu, J., Liu, D., **Guo, Z.**, and Wu, Y.
IEEE Transactions on Visualization and Computer Graphics (Proceedings of IEEE
VIS 2022, 24.9%, 110/441)
- 2021 **TacticFlow: Visual Analytics of Ever-changing Tactics in Racket Sports.**
Wu, J., Liu, D., **Guo, Z.**, Xu, Q., and Wu, Y.
IEEE Transactions on Visualization and Computer Graphics (Proceedings of IEEE
VIS 2021, 25.8%, 114/442)

Workshop Proceedings

- 2024 **Unexploited Information Value in Human-AI Collaboration**
Guo, Z. and Wu, Y. and Hartline, J. D. and Hullman, J.
NeurIPS 2024 Workshop on Behavioral Machine Learning.

Teaching

Teaching Assistantship

- 2025-2027 **Teaching Assistant**, Introduction to Graduate Study
Northwestern University
- 2024-2025 **Teaching Assistant**, Computer Graphics
Northwestern University

2021-2022 **Teaching Assistant**, Computer Architecture
Zhejiang University

Talks

Sep 2026 **ComplLLM: Fine-tuning LLMs to Discover Complementary Signals for Decision-making**
2026 NSF AI-SDM Human-AI Complementarity Workshop
Carnegie Mellon University

Jul 2026 **Measuring and Explaining Complementary Information in Human-AI Decision-making**
2026 ESIF Economics and AI+ML Meeting
Cornell University

Feb 2026 **Complementarity in Human-AI Collaboration**
Invited Speaker at student seminar in Computer Science Department
Northwestern University

Sep 2025 **The Value of Information in Human-AI Decision-making**
2025 NSF AI-SDM Human-AI Complementarity Workshop
Carnegie Mellon University

Service

2026 **Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), Forty-Third International Conference on Machine Learning (ICML), and The Fourteenth International Conference on Learning Representations (ICLR) reviewer**

2025 **The ACM Conference on Human Factors in Computing Systems (CHI), The ACM Conference on Fairness, Accountability, and Transparency (FAccT) reviewer**

2025-present **Northwestern Computer Science Social Initiative board member**

Last updated: September 24, 2026